



Demystify metadata

Now that you understand

the different ways to organize data in a database, let's talk about how you can describe that data. In this video, we'll start

exploring metadata, which is a very important aspect of database management. Metadata is an abstract

concept, though. Let's kick things off with

a simple, everyday example. Did you know that every time a photo is taken

with a smartphone, data is automatically collected and stored within that photo? Take a look.

Choose any

photo on your computer. Here's a cute shot of my

friend's dogs, Rudy and Matilda. On your photo, right-click

on "Get Info" or "Properties." This will give you

the photo's metadata, which may tell you the

type of file it is; the date and time it was taken; the geolocation, or

where it was taken; what kind of device was used to take

the photo; and much more. Pretty amazing, right? Here's

another example. Every time you send

or receive an email, metadata is sent right

along with that message. You can find it by clicking on "Show Original" or "View

Message Details." An email message's metadata

includes its subject, who it's from, who it's to, and the date and time it was sent. The metadata even knows how quickly it was delivered after the sender pressed, "Send." Metadata is information that's used to describe the data that's contained in something, like a photo or an email. Keep in mind that metadata is not the data itself. Instead, it's data about the data. In data analytics, metadata helps data analysts interpret the contents of the data within a database. That's why metadata is so important when working with databases. It tells an analyst what the data is all about. That makes it possible to put the data to work solving problems and making data-driven decisions. As a data analyst, there are three common types of metadata that you'll come across: descriptive, structural, and administrative. Descriptive metadata is metadata that describes a piece of data and can be used to identify it at a later point in time. For instance, the descriptive metadata of a book in a library would include the code you see on its spine, known as a unique International Standard Book Number, also called the ISBN. It would also include the book's author and title. Next is structural metadata, which is metadata that indicates how a piece of data is organized and whether it's part of one or more than one data collection. Let's head back to the library. An example of structural data

would be how the pages of a book are put together to create different chapters. It's important to note that structural metadata also keeps track of the relationship between two things. For example, it can show us that the digital document of a book manuscript was actually the original version of a now printed book. Finally, we have administrative metadata. Administrative metadata is metadata that indicates the technical source of a digital asset. When we looked at the metadata inside the photo, that was administrative metadata. It shows you the type of file it was, the date and time it was taken, and much more. Here's one final thought to help you understand metadata. If you're on your way to the library to pick out a book, you could research a book's title, author, length, and number of chapters. That's all metadata, and it can tell you a lot about the book, but you have to actually read the book to know what it's all about. Likewise, you can read about data analytics, but you have to take this course to earn the Google Data Analytics certificate. Keep moving forward to gain that new perspective.